Week 4

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Where do I get slides/worksheets/solution?

Or go to fredazizi.github.io/Teaching

Quick review (1)

Given a sample space $S = \{O_1, O_2, \ldots, O_k\}$

- $0 \leq P(O_i) \leq 1$ for each i
- $\sum_{i=1}^{k} P(O_i) = 1$.

Quick review (2)

$A \cap B$

- Intersection of Events A and B: the event that occurs when both A and B occur. $A \cup B$
- Union of Events A and B is the event that occurs when either$A$ or B or **both** occur. It is denoted as A or B.
- Conditional Probability: The probability of event A given event B is

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

$\frac{P(A \text{ and } B)}{P(B)} = P(A)$

- Independent Events: A and B are said to be independent if $P(A \mid B) = P(A)$ or $P(B \mid A) = P(B)$. $P(A \cap B) = P(A) \cdot P(B)$
- Mutually exclusive: no two outcomes can occur at the same time.

$P(A \cap B) = 0$

Quick review (3)

A not happening

- Complement Rule: $P\left(A^C\right) = 1 - P(A)$.
- Multiplication Rule: $P(A \text{ and } B) = P(B)P(A \mid B)$.
- Addition Rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

  $P(A \cup B)$ $A \cap B$
- Bayes Rule:
  $P(A_i \mid B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1)+P(A_2)P(B|A_2)+\cdots+P(A_k)P(B|A_k)}$